

Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r} a^r b^{n-r}$$

Multinomial

$$\binom{n}{r} \binom{n-r}{s} a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Classified Practice

Binomial and Multinomial Theorems

Q001 Challenge

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Choose the correct option.

- A Coefficient = 720; sum = -1
- B Coefficient = 720; sum = 1
- C Coefficient = 540; sum = -1
- D Coefficient = 240; sum = 32

Classified Practice

Binomial and Multinomial Theorems

Q002 Challenge

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

Choose the correct option.

- A First = 60; second = -120
- B First = 60; second = 120
- C First = 120; second = -60
- D First = -60; second = -120

Classified Practice

Binomial and Multinomial Theorems

Q003 Challenge

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

Choose the correct option.

- A Greatest coefficient = 19
- B Greatest coefficient = 16
- C Greatest coefficient = 20
- D Greatest coefficient = 24

Classified Practice

Binomial and Multinomial Theorems

Q004 **Written****Expand $(x + 2)^5$ using the Binomial Theorem.**

Q005 MCQ**Expand $(3x - 2)^4$ using the Binomial Theorem:****Choose the correct option.**

A $81x^4 - 216x^3 - 96x + 16$

B $81x^4 - 216x^3 + 432x^2 - 96x + 16$

C $-(3x - 2)^4$

D $81x^4 - 216x^3 + 216x^2 - 96x + 16$

Classified Practice

Binomial and Multinomial Theorems

Q006 Written

Expand $(x + 2)^6$ using the Binomial Theorem.

Q007 MCQ**Expand $(x + 1)^4$ using the Binomial Theorem:****Choose the correct option.**

A $x^4 + 4x^3 + 4x + 1$

B $x^4 + 4x^3 + 6x^2 + 4x + 1$

C $x^4 + 4x^3 + 12x^2 + 4x + 1$

D $-(x + 1)^4$

Classified Practice

Binomial and Multinomial Theorems

Q008 MCQ

Expand $(a + b)^4$ using the Binomial Theorem:**Choose the correct option.**

A $a^4 + 4a^3b + 4ab^3 + b^4$

B $a^4 + 4a^3b + 12a^2b^2 + 4ab^3 + b^4$

C $a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$

D $-(a + b)^4$

Q009 Written**Expand $(a - 2b)^4$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q010 MCQ

Expand $(x - 2)^5$ using the Binomial Theorem:**Choose the correct option.**

A $x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$

B $x^5 - 10x^4 + 40x^3 - 80x^2 - 32$

C $x^5 - 10x^4 + 40x^3 - 80x^2 + 160x - 32$

D $-(x - 2)^5$

Q011 Written**Expand $(3x + 2y)^5$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.

Q013 Written**Expand $(x - 3)^6$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q014 Written

Expand $(2x - 3y)^6$ using the Binomial Theorem.

Q015 Written**Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q016 MCQ

Expand $(x + 3)^3$ using the Binomial Theorem:**Choose the correct option.**

A $x^3 + 9x^2 + 27$

B $x^3 + 9x^2 + 27x + 27$

C $x^3 + 9x^2 + 54x + 27$

D $-(x + 3)^3$

Classified Practice

Binomial and Multinomial Theorems

Q017 MCQ

Expand $(a + 2b)^5$ using the Binomial Theorem:**Choose the correct option.**

A $a^5 + 10a^4b + 40a^3b^2 + 80ab^4 + 32b^5$

B $a^5 + 10a^4b + 40a^3b^2 + 160a^2b^3 + 80ab^4 + 32b^5$

C $-(a + 2b)^5$

D $a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$

Classified Practice

Binomial and Multinomial Theorems

Q018 MCQ

Expand $(x + 2)^2$ using the Binomial Theorem:**Choose the correct option.**

A $x^2 + 4$

B $x^2 + 8x + 4$

C $-(x + 2)^2$

D $x^2 + 4x + 4$

Q019 MCQ

Expand $(a - b)^3$ using the Binomial Theorem:**Choose the correct option.**

A $a^3 - 3a^2b + 3ab^2 - b^3$

B $a^3 - 3a^2b - b^3$

C $a^3 - 3a^2b + 6ab^2 - b^3$

D $-(a - b)^3$

Classified Practice

Binomial and Multinomial Theorems

Q020 MCQ

Expand $(2x + 1)^5$ using the Binomial Theorem:**Choose the correct option.**

A $32x^5 + 80x^3 + 40x^2 + 10x + 1$

B $32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$

C $32x^5 + 160x^4 + 80x^3 + 40x^2 + 10x + 1$

D $-(2x + 1)^5$

Q021 Written**Expand $(3a - 2b)^5$ using the Binomial Theorem.**

Classified Practice

Binomial and Multinomial Theorems

Q022 MCQ

Expand $(2x - 1)^4$ using the Binomial Theorem:**Choose the correct option.**

A $16x^4 - 32x^3 - 8x + 1$

B $16x^4 - 32x^3 + 48x^2 - 8x + 1$

C $16x^4 - 32x^3 + 24x^2 - 8x + 1$

D $-(2x - 1)^4$

Q023 MCQ**Expand $(x - 3)^4$ using the Binomial Theorem:****Choose the correct option.**

A $x^4 - 12x^3 - 108x + 81$

B $x^4 - 12x^3 + 54x^2 - 108x + 81$

C $x^4 - 12x^3 + 108x^2 - 108x + 81$

D $-(x - 3)^4$

Classified Practice

Binomial and Multinomial Theorems

Q024 MCQ

Expand $(2a + b)^3$ using the Binomial Theorem:**Choose the correct option.**

A $8a^3 + 6ab^2 + b^3$

B $8a^3 + 24a^2b + 6ab^2 + b^3$

C $8a^3 + 12a^2b + 6ab^2 + b^3$

D $-(2a + b)^3$

Q025 MCQ

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:**Choose the correct option.**

A $16a^8 + 96a^6 + 216a^2 + 81$

B $16a^8 + 96a^6 + 432a^4 + 216a^2 + 81$

C $16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$

D $-(2a^2 + 3)^4$

Classified Practice

Binomial and Multinomial Theorems

Q026 MCQ

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:**Choose the correct option.**

A $27a^6 - 54a^4 - 8$

B $27a^6 - 54a^4 + 36a^2 - 8$

C $27a^6 - 54a^4 + 72a^2 - 8$

D $-(3a^2 - 2)^3$

Q027 Written**Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.**

Classified Practice

Binomial and Multinomial Theorems

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.

Q029 Written**Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.**

Classified Practice

Binomial and Multinomial Theorems

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.

Q031 MCQ**Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:****Choose the correct option.**

- A 720
 - B -720
 - C -1440
 - D -719
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q032 Written

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.

Q033 MCQ**Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:****Choose the correct option.**

- A 1120
 - B -1120
 - C 2240
 - D 1121
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:

Choose the correct option.

A -12

B 12

C 24

D 13

Q035 MCQ**Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:****Choose the correct option.**A -160 B 320 C 160 D 161

Classified Practice

Binomial and Multinomial Theorems

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:

Choose the correct option.

A -405

B 810

C 406

D 405

Q037 Written**Find the coefficient of a in the expansion of $(2a - 3)^4$.**

Classified Practice

Binomial and Multinomial Theorems

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.

Q039 Written**Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.**

Classified Practice

Binomial and Multinomial Theorems

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:

Choose the correct option.

- A 90
- B -180
- C -89
- D -90

Q041 Written**Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.**

Classified Practice

Binomial and Multinomial Theorems

Q042 Written

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

Q043 MCQ**Find the coefficient of xyz in the expansion of $(x + y + z)^3$ ∴****Choose the correct option.**

- A 6
 - B -6
 - C 3
 - D 12
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:

Choose the correct option.

A -280

B 560

C 280

D 281

Q045 MCQ**Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:****Choose the correct option.**

A 60

B -60

C 120

D 61

Classified Practice

Binomial and Multinomial Theorems

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:

Choose the correct option.

A -90

B 90

C 180

D 91

Q047 MCQ**Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:****Choose the correct option.****A 42****B -42** **C 84****D 43**

Classified Practice

Binomial and Multinomial Theorems

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.

Q049 Written**Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.**

Classified Practice

Binomial and Multinomial Theorems

Q050 Written

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.

Q051 Written**Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.**

Classified Practice

Binomial and Multinomial Theorems

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.

Q053 MCQ**Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:****Choose the correct option.**

- A 6720
 - B -13440
 - C -6719
 - D -6720
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:

Choose the correct option.

A -3240

B 6480

C 3240

D 3241

Q055 MCQ**Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:****Choose the correct option.**

- A 4320
 - B -8640
 - C -4319
 - D -4320
-
-
-

Classified Practice

Binomial and Multinomial Theorems

Q056 Written

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.

Q057 Written**Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.**

Classified Practice

Binomial and Multinomial Theorems

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.

Classified Practice

Binomial and Multinomial Theorems

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.

Classified Practice

Binomial and Multinomial Theorems

Q062 Written

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Choose the correct option.

A -240

B 240

C -120

D 120

Classified Practice

Binomial and Multinomial Theorems

Quiz MCQ 2 **Concept Quiz · MCQ**

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

Choose the correct option.

A 30

B 60

C 90

D 120

Quiz MCQ 3 Concept Quiz · MCQ

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

Choose the correct option.

- A -4320
- B -2160
- C 2160
- D 4320

Classified Practice

Binomial and Multinomial Theorems

Quiz MCQ 4 **Concept Quiz · MCQ**

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

Choose the correct option.

A 0

B 8

C 16

D 32

Quiz WRITTEN 5 Concept Quiz · Written**Expand $(2x - 1)^4$ fully.**

Classified Practice

Binomial and Multinomial Theorems

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

Classified Practice

Binomial and Multinomial Theorems

Quiz WRITTEN 8 Concept Quiz · Written

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Answer key

Use the question number shown on each page.

Q001 A	Q018 D	Q036 D	Q054 C
Q002 A	Q019 A	Q037 -216	Q055 D
Q003 A	Q020 B	Q038 1080	Q056 15120
Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$	Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$	Q039 -160	Q057 -20412
Q005 D	Q022 C	Q040 D	Q058 -5040
Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$	Q023 B	Q041 4320	Q059 136080
Q007 B	Q024 C	Q042 32	Q060 -120960
Q008 C	Q025 C	Q043 A	Q061 -40320
Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$	Q026 B	Q044 C	Q062 19
Q010 A	Q027 10	Q045 A	Quiz MCQ 1 A
Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$	Q028 60	Q046 B	Quiz MCQ 2 C
Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$	Q029 -20412	Q047 A	Quiz MCQ 3 A
Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$	Q030 -1344	Q048 105	Quiz MCQ 4 A
Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$	Q031 B	Q049 210	Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$
Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$	Q032 -48384	Q050 252	Quiz WRITTEN 6 720
Q016 B	Q033 A	Q051 1680	Quiz WRITTEN 7 -480
Q017 D	Q034 B	Q052 1260	Quiz WRITTEN 8 19
	Q035 C	Q053 D	

Concept 2 | Polynomial Division

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Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Classified Practice

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Choose the correct option.

A $Q = 2x - 2, \quad R = -2x + 2$

B $Q = 2x - 1, \quad R = -2x + 1$

C $Q = 2x - 1, \quad R = 2x - 1$

D $Q = 2x, \quad R = -2x + 1$

Classified Practice

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Choose the correct option.

A $3x^3 - 7x^2 + 12x - 1$

B $(3x - 1) \times (x^2 - 2x + 3)$

C $3x^3 - 7x^2 + 10x - 5$

D $x^2 + 2x + 4$

Classified Practice

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Choose the correct option.

A $\frac{x^3+5x^2+9x-11}{x+3}$

B $\frac{-x^2-2x+1}{x^3+5x^2+5x-3}$

C $\frac{x^3+5x^2+5x-3}{x+4}$

D $x^2 + 2x - 1$

Classified Practice

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Choose the correct option.

A $2x^3 + 10x^2 + 6x - 18$

B $2x + 6$

C $2x + 5$

D $\frac{1}{2x+5}$

Classified Practice

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Choose the correct option.

A $Q = 3x^2 + 5x + 5, \quad R = -2x - 10$

B $Q = 3x^2 + 5x + 6, \quad R = -2x - 11$

C $Q = 3x^2 + 5x + 6, \quad R = 2x + 11$

D $Q = 3x^2 + 5x + 7, \quad R = -2x - 11$

Classified Practice

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Choose the correct option.

A $Q = x \times (x^2 - 2x + 1), \quad R = 6$

B $Q = x \times (x^2 - 2x + 1), \quad R = -6$

C $Q = x^3 - 2x^2 + x + 1, \quad R = 6$

D $Q = x^3 - 2x^2 + x - 1, \quad R = 7$

Classified Practice

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Q008 Written**Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.**

Classified Practice

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:

Choose the correct option.

A $Q = 3x^2 + 8x + 9, \quad R = -23$

B $Q = 3x^2 + 8x + 10, \quad R = 23$

C $Q = 3x^2 + 8x + 8, \quad R = 24$

D $Q = 3x^2 + 8x + 9, \quad R = 23$

Classified Practice

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:

Choose the correct option.

A $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{37}{8}, R = \frac{111}{8}$

B $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = \frac{103}{8}$

C $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, R = -\frac{103}{8}$

D $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{53}{8}, R = \frac{103}{8}$

Classified Practice

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:

Choose the correct option.

A $Q = x \times (3x - 1), \quad R = 9$

B $Q = 3x^2 - x + 1, \quad R = 8$

C $Q = 3x^2 - x + 1, \quad R = -8$

D $Q = 3x^2 - x + 2, \quad R = 8$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:

Choose the correct option.

- A $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$
- B $Q = 2x^3 - 3x^2 + 6x - 17, \quad R = -38$
- C $Q = 2x^3 - 3x^2 + 6x - 16, \quad R = 38$
- D $Q = 2x^3 - 3x^2 + 6x - 18, \quad R = 39$

Classified Practice

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:

Choose the correct option.

A $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = -36$

B $Q = 4x^3 + 5x^2 + 10x + 22, \quad R = 36$

C $Q = 4x^3 + 5x^2 + 10x + 20, \quad R = 37$

D $Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$

Classified Practice

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:

Choose the correct option.

A $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{100}{27}, \quad R = -\frac{92}{27}$

B $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$

C $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = \frac{119}{27}$

D $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{46}{27}, \quad R = -\frac{119}{27}$

Classified Practice

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Q016 Written**Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.**

Classified Practice

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Q018 Written**Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.**

Classified Practice

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:

Choose the correct option.

A $Q = x + 1, \quad R = 4x - 6$

B $Q = x + 2, \quad R = -4x + 6$

C $Q = x, \quad R = -4x + 7$

D $Q = x + 1, \quad R = -4x + 6$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:

Choose the correct option.

A $Q = x^2 - 5x + 10, \quad R = -25x + 7$

B $Q = x^2 - 5x + 11, \quad R = -25x + 6$

C $Q = x^2 - 5x + 11, \quad R = 25x - 6$

D $Q = x^2 - 5x + 12, \quad R = -25x + 6$

Classified Practice

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:

Choose the correct option.

A $Q = 2x + 5, \quad R = -2x + 5$

B $Q = 2x + 6, \quad R = 2x - 5$

C $Q = 2x + 4, \quad R = 2x - 4$

D $Q = 2x + 5, \quad R = 2x - 5$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:

Choose the correct option.

A $Q = x^2 - 5x + 22, \quad R = -69x + 24$

B $Q = x^2 - 5x + 20, \quad R = -69x + 25$

C $Q = x^2 - 5x + 21, \quad R = -69x + 24$

D $Q = x^2 - 5x + 21, \quad R = 69x - 24$

Classified Practice

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:

Choose the correct option.

A $Q = 3x + 2, \quad R = x + 6$

B $Q = 3x + 3, \quad R = x + 5$

C $Q = 3x + 3, \quad R = -x - 5$

D $Q = 3x + 4, \quad R = x + 5$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:

Choose the correct option.

A $Q = 2x^2 - 3x + 1, \quad R = 8x - 10$

B $Q = 2x^2 - 3x + 1, \quad R = -8x + 10$

C $Q = 2x^2 - 3x + 2, \quad R = 8x - 10$

D $Q = x \times (2x - 3), \quad R = 8x - 9$

Classified Practice

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Q026 Written

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

Classified Practice

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:

Choose the correct option.

A $Q = 2x - \frac{3}{2}, R = -\frac{35x}{2} + 9$

B $Q = 2x - \frac{1}{2}, R = \frac{35x}{2} - 9$

C $Q = 2x - \frac{5}{2}, R = \frac{35x}{2} - 8$

D $Q = 2x - \frac{3}{2}, R = \frac{35x}{2} - 9$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:

Choose the correct option.

A $Q = x - 1, \quad R = -4x + 3$

B $Q = x, \quad R = -4x + 2$

C $Q = x, \quad R = 4x - 2$

D $Q = x + 1, \quad R = -4x + 2$

Classified Practice

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:

Choose the correct option.

- A $Q = \frac{5x}{2} - \frac{3}{4}, R = \frac{9x}{4} + \frac{21}{4}$
- B $Q = \frac{5x}{2} + \frac{1}{4}, R = -\frac{9x}{4} - \frac{21}{4}$
- C $Q = \frac{5x}{2} - \frac{7}{4}, R = -\frac{9x}{4} - \frac{17}{4}$
- D $Q = \frac{5x}{2} - \frac{3}{4}, R = -\frac{9x}{4} - \frac{21}{4}$

Classified Practice

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:

Choose the correct option.

- A $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{67}{64}, R = \frac{37x}{64} + \frac{163}{32}$
- B $Q = \frac{3x^2}{4} + \frac{5x}{16} - \frac{61}{64}, R = \frac{37x}{64} + \frac{195}{32}$
- C $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = \frac{37x}{64} + \frac{163}{32}$
- D $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, R = -\frac{37x}{64} - \frac{163}{32}$

Classified Practice

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:

Choose the correct option.

- A $Q = \frac{3x}{2} + \frac{11}{4}, R = \frac{11x}{4} - \frac{3}{4}$
- B $Q = \frac{3x}{2} + \frac{15}{4}, R = -\frac{11x}{4} + \frac{3}{4}$
- C $Q = \frac{3x}{2} + \frac{7}{4}, R = -\frac{11x}{4} + \frac{7}{4}$
- D $Q = \frac{3x}{2} + \frac{11}{4}, R = -\frac{11x}{4} + \frac{3}{4}$

Classified Practice

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Choose the correct option.

A $Q = 3x - \frac{11}{2}, R = \frac{29x}{2} - 7$

B $Q = 3x - \frac{9}{2}, R = \frac{29x}{2} - 8$

C $Q = 3x - \frac{9}{2}, R = -\frac{29x}{2} + 8$

D $Q = 3x - \frac{7}{2}, R = \frac{29x}{2} - 8$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Classified Practice

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Choose the correct option.

A $Q = 4a^2 - 3ax + x^2, \quad R = -2a^3$

B $Q = 4a^2 - 3ax + x^2 + 1, \quad R = 2a^3$

C $Q = 4a^2 - 3ax + x^2 - 1, \quad R = 2a^3 + 1$

D $Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Choose the correct option.

A $Q = x + 2y - 1, \quad R = -6y^2 + 1$

B $Q = x + 2y, \quad R = -6y^2$

C $Q = x + 2y, \quad R = 6y^2$

D $Q = x + 2y + 1, \quad R = -6y^2$

Classified Practice

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Classified Practice

Polynomial Division

Q042 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Q043 Written

Find the polynomial A when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Classified Practice

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Classified Practice

Polynomial Division

Q046 Written

When $4x^3 + 8x - 9$ is divided by B , the quotient is $2x - 3$ and the remainder is $x + 15$. Find B .

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B , the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B .

Classified Practice

Polynomial Division

Q048 **Written**

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B , the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B .

Classified Practice

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B , the quotient is $x + 2$ and the remainder is $3x + 13$. Find B .

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B , the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B .

Classified Practice

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R?:

Choose the correct option.

A -5

B -1

C 5

D 1

Classified Practice

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Choose the correct option.

A *another variable to divide by*

B a constant coefficient

C *zero*

D the divisor

Quiz MCQ 3 Concept Quiz · MCQ

To recover **B** from $A = B \times Q + R$, use... Which of the following is correct?:

Choose the correct option.

A $B = A \times Q + R$

B $B = \frac{A+R}{Q}$

C $B = \frac{A-R}{Q}$

D $B = \frac{Q}{A-R}$

Classified Practice

Polynomial Division

Quiz MCQ 4 **Concept Quiz · MCQ**

If B has degree 2, a valid remainder can have degree...:

Choose the correct option.

A 3

B 2

C 1

D 4

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q, R, and verify $A = B \times Q + R$.

Classified Practice

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written**Recover A when $B=x^2 - x - 2$, $Q=2x + 3$, $R=-x + 1$.**

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Classified Practice

Polynomial Division

Quiz WRITTEN 8 **Concept Quiz · Written**

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4, R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1, R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2, R = 0$	Q033 $Q = 3a + 4x, R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2, R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x, R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2, R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3, R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2, R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5, R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1, R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2, R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18, R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4, R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2, R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2, R = 2x + 1$

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Classified Practice

Rational Expressions

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for 2():

Choose the correct option.

A $\frac{2x(x+3)}{(x-3)(x-1)}$

B $\frac{-2x^2-6x}{x^2-4x+3}$

C $\frac{4x^2+12x}{x^2-4x+3}$

D $\frac{x^2+3x}{x^2-4x+3}$

Classified Practice

Rational Expressions

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3():

Choose the correct option.

A $\frac{-3x^2+3x+18}{x^2+3x}$

B $\frac{3(x-3)(x+2)}{x(x+3)}$

C $\frac{6x^2-6x-36}{x^2+3x}$

D $\frac{3x^2-3x-18}{2x^2+6x}$

Classified Practice

Rational Expressions

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for 4():

Choose the correct option.

A $\frac{-8x^2+24x+32}{x^2-5x+6}$

B $\frac{16x^2-48x-64}{x^2-5x+6}$

C $\frac{8(x-4)(x+1)}{(x-3)(x-2)}$

D $\frac{4x^2-12x-16}{x^2-5x+6}$

Classified Practice

Rational Expressions

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for 8():

Choose the correct option.

A $-\frac{8}{x}$

B $\frac{16}{x}$

C $-\frac{x}{4}$

D $-\frac{8}{x}$

Classified Practice

Rational Expressions

Q005 Challenge

After simplifying this transfer chain before substitution, which result is correct for 9():

Choose the correct option.

A $\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$

B $-\frac{9x^5}{2} + \frac{9}{2}$

C $9x^5 - 9$

D $\frac{9x^5}{4} - \frac{9}{4}$

Classified Practice

Rational Expressions

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for 10 ()?:

Choose the correct option.

A $\frac{-10x^3+10x^2+160x-160}{x^3-5x^2+6x}$

B $\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$

C $\frac{20x^3-20x^2-320x+320}{x^3-5x^2+6x}$

D $\frac{5x^3-5x^2-80x+80}{x^3-5x^2+6x}$

Classified Practice

Rational Expressions

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.

Q008 Written**Simplify** $\frac{6x^2y^3}{20x^3y}$.

Classified Practice

Rational Expressions

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.

Q010 Written**Simplify** $\frac{2a^2b}{8a^3b^3}$.

Classified Practice

Rational Expressions

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Choose the correct option.

A $-\frac{4b^3}{a^2}$

B $\frac{4b^3}{a^2}$

C $\frac{8b^3}{a^2}$

D $\frac{2b^3}{a^2}$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Choose the correct option.

- A $\frac{9y^2}{10x}$
B $-\frac{9y^2}{10x}$
C $\frac{9y^2}{5x}$
D $\frac{5x}{9y^2}$

Classified Practice

Rational Expressions

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Choose the correct option.

- A $\frac{3a}{b^3}$
B $-\frac{3a}{b^3}$
C $\frac{6a}{b^3}$
D $\frac{3a}{2b^3}$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Choose the correct option.

A $-\frac{1}{2ab^2}$

B $\frac{1}{2ab^2}$

C $\frac{1}{ab^2}$

D $\frac{1}{4ab^2}$

Classified Practice

Rational Expressions

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Choose the correct option.

A $\frac{3a^2}{2b^3}$

B $-\frac{3a^2}{2b^3}$

C $\frac{3a^2}{b^3}$

D $\frac{3a^2}{4b^3}$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Choose the correct option.

A $\frac{9x^2y^2}{4}$

B $-\frac{4}{9x^2y^2}$

C $-\frac{4}{9x^2y^2}$

D $-\frac{2}{8x^2y^2}$

Classified Practice

Rational Expressions

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Choose the correct option.

A $-\frac{3b^3}{a^3c}$

B $\frac{6b^3}{a^3c}$

C $\frac{3b^3}{a^3c}$

D $\frac{3b^3}{2a^3c}$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Choose the correct option.

A $-\frac{4x^3}{3y^2}$

B $\frac{8x^3}{3y^2}$

C $\frac{2x^3}{3y^2}$

D $\frac{4x^3}{3y^2}$

Classified Practice

Rational Expressions

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Choose the correct option.

A $\frac{2a^3}{b^2}$

B $-\frac{2a^3}{b^2}$

C $\frac{4a^3}{b^2}$

D $\frac{a^3}{b^2}$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Choose the correct option.

A $-3x^2y^3$

B $3x^2y^3$

C $6x^2y^3$

D $\frac{3x^2y^3}{2}$

Classified Practice

Rational Expressions

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Choose the correct option.

A $\frac{3c^2}{2a^3}$

B $-\frac{3c^2}{a^3}$

C $-\frac{3c^2}{2a^3}$

D $-\frac{3c^2}{4a^3}$

Q022 Written**Simplify** $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Classified Practice

Rational Expressions

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Choose the correct option.

A $-\frac{12c^2}{a}$

B $\frac{24c^2}{a}$

C $\frac{12c^2}{a}$

D $\frac{6c^2}{a}$

Classified Practice

Rational Expressions

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Choose the correct option.

A $-\frac{8a^2c^3}{b^2}$

B $\frac{16a^2c^3}{b^2}$

C $\frac{4a^2c^3}{b^2}$

D $\frac{8a^2c^3}{b^2}$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3 \left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3} \right)$?:

Choose the correct option.

- A $-\frac{27ac}{2b^2}$
- B $\frac{27ac}{b^2}$
- C $\frac{27ac}{4b^2}$
- D $\frac{4b^2}{27ac}$

Classified Practice

Rational Expressions

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Choose the correct option.

A $\frac{9ax^2}{y}$

B $-\frac{9ax^2}{y}$

C $\frac{18ax^2}{y}$

D $\frac{9ax^2}{2y}$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Choose the correct option.

A $-\frac{64bc}{5a}$

B $\frac{64bc}{5a}$

C $\frac{5a}{128bc}$

D $\frac{5a}{32bc}$

Classified Practice

Rational Expressions

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Choose the correct option.

A $-\frac{18abxy^2}{5}$

B $\frac{36abxy^2}{5}$

C $\frac{18abxy^2}{5}$

D $\frac{9abxy^2}{5}$

Q030 Written**Simplify** $\frac{x^2-4}{3x^2+7x+2}$.

Classified Practice

Rational Expressions

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.

Q032 Written**Simplify** $\frac{x^2+3x}{x^2-9}$.

Classified Practice

Rational Expressions

Q033 Written

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Q034 Written**Simplify** $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Classified Practice

Rational Expressions

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Classified Practice

Rational Expressions

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}$.

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}$.

Classified Practice

Rational Expressions

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Classified Practice

Rational Expressions

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Classified Practice

Rational Expressions

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Classified Practice

Rational Expressions

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ **for** $x = \sqrt{123}$.

Classified Practice

Rational Expressions

Q047 **Challenge****Simplify** $\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right)$; state every excluded value.

Classified Practice

Rational Expressions

Q048 Written

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

Classified Practice

Rational Expressions

Q049 MCQ

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1}:$$

Choose the correct option.

A $\frac{x^2 + 5x + 3}{x + 1}$

B $\frac{x^2 + 3x + 1}{x + 1}$

C $\frac{x^2 + 4x + 2}{x + 1}$

D $-\frac{x^2 + 4x + 2}{x + 1}$

Classified Practice

Rational Expressions

Q050 MCQ

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4}:$$

Choose the correct option.

A $\frac{3x^2 + x - 3}{(x - 2)(x + 2)}$

B $\frac{x^2 + x + 5}{(x - 2)(x + 2)}$

C $-\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$

D $\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$

Classified Practice

Rational Expressions

Q051 MCQ

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)}:$$

Choose the correct option.

A $\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$

B $\frac{4x^2 + 4x - 7}{(x - 2)(x + 1)}$

C $\frac{2x^2 + 6x - 3}{(x - 2)(x + 1)}$

D $-\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$

Classified Practice

Rational Expressions

Q052 MCQ

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2}:$$

Choose the correct option.

A $\frac{x^3 + 2x^2 + 2x + 5}{x + 2}$

B $\frac{x^3 + 2x^2 + x + 3}{x + 2}$

C $\frac{x^3 + 2x^2 + 1}{x + 2}$

D $-\frac{x^3 + 2x^2 + x + 3}{x + 2}$

Classified Practice

Rational Expressions

Q053 Written

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

Classified Practice

Rational Expressions

Q054 MCQ

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2}:$$

Choose the correct option.

A $\frac{x(x + 3)}{x + 2}$

B $\frac{x^2 + 4x + 2}{x + 2}$

C $\frac{x^2 + 2x - 2}{x + 2}$

D $-\frac{x(x + 3)}{x + 2}$

Classified Practice

Rational Expressions

Q055 MCQ

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1}:$$

Choose the correct option.

A $\frac{4x^2 - 4x + 5}{(x - 1)(x + 1)}$

B $\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$

C $\frac{2x^2 - 4x + 7}{(x - 1)(x + 1)}$

D $-\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$

Classified Practice

Rational Expressions

Q056 MCQ

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)}:$$

Choose the correct option.

A $\frac{x^2 - 2x + 4}{(x - 1)(x + 2)}$

B $-\frac{x^2 + 4x - 8}{(x - 1)(x + 2)}$

C $-\frac{3(x - 2)}{(x - 1)(x + 2)}$

D $\frac{3(x - 2)}{(x - 1)(x + 2)}$

Classified Practice

Rational Expressions

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Choose the correct option.

A $\frac{2x^3 - x^2 + 2x - 10}{x - 3}$

B $\frac{2x^3 - x^2 - 4}{x - 3}$

C $\frac{2x^3 - x^2 + x - 7}{x - 3}$

D $-\frac{2x^3 - x^2 + x - 7}{x - 3}$

Classified Practice

Rational Expressions

Q058 Written

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}$$

Q059 Written

Simplify

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

Classified Practice

Rational Expressions

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)}:$$

Choose the correct option.

A $\frac{x^3 + 12x^2 + 30x + 11}{(x+1)(x+3)(x+5)}$

B $\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$

C $-\frac{x^3 + 6x^2 + 16x + 19}{(x+1)(x+3)(x+5)}$

D $-\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$

Classified Practice

Rational Expressions

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)}:$$

Choose the correct option.

- A $\frac{x^3 + 14x^2 + 53x + 62}{(x+2)(x+4)(x+6)}$
- B $-\frac{x^3 + 10x^2 + 35x + 34}{(x+2)(x+4)(x+6)}$
- C $\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$
- D $-\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$

Classified Practice

Rational Expressions

Q062 MCQ

Simplify

$$\frac{3x + 2}{(x + 1)(x + 4)} + \frac{x + 5}{(x + 4)(x + 7)}:$$

Choose the correct option.

A $\frac{x^3 + 16x^2 + 68x + 47}{(x + 1)(x + 4)(x + 7)}$

B $-\frac{x^3 + 8x^2 + 10x + 9}{(x + 1)(x + 4)(x + 7)}$

C $-\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$

D $\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$

Classified Practice

Rational Expressions

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)}:$$

Choose the correct option.

- A $\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$
- B $\frac{x^3 + 20x^2 + 80x + 82}{(x+2)(x+5)(x+8)}$
- C $-\frac{x^3 + 10x^2 + 52x + 78}{(x+2)(x+5)(x+8)}$
- D $-\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$

Classified Practice

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x + 1}{(x + 3)(x + 6)} + \frac{x + 4}{(x + 6)(x + 9)}:$$

Choose the correct option.

A $\frac{x^3 + 21x^2 + 125x + 183}{(x + 3)(x + 6)(x + 9)}$

B $\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$

C $-\frac{x^3 + 15x^2 + 73x + 141}{(x + 3)(x + 6)(x + 9)}$

D $-\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$

Classified Practice

Rational Expressions

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)}:$$

Choose the correct option.

- A $\frac{x^3 + 18x^2 + 63x + 25}{(x+1)(x+5)(x+8)}$
- B $-\frac{x^3 + 10x^2 + 43x + 55}{(x+1)(x+5)(x+8)}$
- C $\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$
- D $-\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$

Classified Practice

Rational Expressions

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Q067 Written**Simplify**

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Classified Practice

Rational Expressions

Q068 Written

Simplify

$$\frac{2}{x-1} + \frac{3}{x^2+5x-6}.$$

Classified Practice

Rational Expressions

Q069 MCQ

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1};$$

Choose the correct option.

A $\frac{x^2 + 3x - 1}{(x+1)(x+3)}$

B $-\frac{x^2 + 5x + 7}{(x+1)(x+3)}$

C $-\frac{x+4}{(x+1)(x+3)}$

D $\frac{x+4}{(x+1)(x+3)}$

Classified Practice

Rational Expressions

Q070 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

Q071 **Written****Simplify**

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}$$

Classified Practice

Rational Expressions

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}$$

Q073 Written**Simplify**

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

Classified Practice

Rational Expressions

Q074 Written

Simplify

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

Q075 Written

Simplify

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

Classified Practice

Rational Expressions

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}$$

Classified Practice

Rational Expressions

Q077 MCQ

A learner combines

$$\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$$

Correct the numerator line, then simplify the expression:

Choose the correct option.

A
$$\frac{1}{(x-1)(x+2)}$$

B
$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

C
$$-\frac{2x^2+2x-5}{(x-1)(x+2)}$$

D
$$\frac{x^2+x-3}{(x-1)(x+2)}$$

Classified Practice

Rational Expressions

Q078 Written

Simplify

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

Classified Practice

Rational Expressions

Q079 **Challenge**

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Classified Practice

Rational Expressions

Q080 Written

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

Q081 Written**Simplify**

$$\frac{1}{1 - \frac{x}{x+1}}.$$

Classified Practice

Rational Expressions

Q082 MCQ

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}}:$$

Choose the correct option.

A $-x$ B $-x - 2$ C $x + 1$ D $-x - 1$

Q083 MCQ

Simplify

$$\frac{1}{1 - \frac{x+3}{x+2}}:$$

Choose the correct option.

A $-x - 2$

B $-x - 1$

C $-x - 3$

D $x + 2$

Classified Practice

Rational Expressions

Q084 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}}:$$

Choose the correct option.

A $-\frac{1}{x}$

B $-\frac{x}{x+1}$

C $-\frac{x}{2x+1}$

D $\frac{x+1}{x}$

Q085 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+3}{x+1}}:$$

Choose the correct option.

- A $\frac{1}{x+2}$
B $-\frac{2x+3}{x+2}$
C $-\frac{x+1}{x+2}$
D $\frac{x+1}{x+2}$

Classified Practice

Rational Expressions

Q086 MCQ

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}}:$$

Choose the correct option.

- A $\frac{x-3}{2x-1}$
- B $-\frac{3x+1}{2x-1}$
- C $\frac{x+2}{2x-1}$
- D $-\frac{x+2}{2x-1}$

Q087 MCQ

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}}:$$

Choose the correct option.

A $-x - 3$

B $-x - 2$

C $-x - 4$

D $x + 3$

Classified Practice

Rational Expressions

Q088 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}}:$$

Choose the correct option.

- A $\frac{3}{x+4}$
- B $-\frac{x+1}{x+4}$
- C $-\frac{2x+5}{x+4}$
- D $\frac{x+1}{x+4}$

Q089 Written**Simplify**

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

Classified Practice

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

Choose the correct option.

- A $\frac{2x+5}{3}$
B $\frac{2(x+4)}{3}$
C $\frac{2(x+1)}{3}$
D $-\frac{2x+5}{3}$

Q091 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

Choose the correct option.

A $\frac{x+3}{2}$

B $\frac{x+5}{2}$

C $\frac{x+1}{2}$

D $-\frac{x+3}{2}$

Classified Practice

Rational Expressions

Q092 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Q093 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

Choose the correct option.

A 2

B 0

C 1

D -1

Classified Practice

Rational Expressions

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

Choose the correct option.

A 2

B 0

C -1

D 1

Q095 MCQ

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

Choose the correct option.

A 2

B 1

C 0

D -1

Classified Practice

Rational Expressions

Q096 Written

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

Q097 **Written****Simplify**

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

Classified Practice

Rational Expressions

Q098 Written

Simplify

$$\frac{\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}}$$

Q099 Written**Simplify**

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

Classified Practice

Rational Expressions

Q100 Written

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

Q101 Written**Simplify**

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

Classified Practice

Rational Expressions

Q102 Written

Simplify

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + x}}}$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Choose the correct option.

- A $\frac{x-2}{3x+1}$
B $\frac{3x+1}{x+2}$
C $\frac{3x+1}{x-2}$
D $\frac{3x+2}{x^2-4}$

Classified Practice

Rational Expressions

Quiz MCQ 2 Concept Quiz · MCQ

Simplify

$$\frac{x+1}{x-2} + \frac{2}{x-2}:$$

Choose the correct option.

- A $\frac{x+3}{x-2}$
- B $\frac{x-1}{x-2}$
- C $\frac{x+1}{2x-4}$
- D $\frac{x+3}{x+2}$

Quiz MCQ 3 Concept Quiz · MCQ

In $\frac{3}{x^2 + x} - \frac{3 - x}{x^2 + x}$, the numerator after the minus sign is... Which of the following is correct?:

Choose the correct option.

A $3 - (3 - x)$

B $3 - 3 - x$

C $3 + 3 - x$

D $3 - x$

Classified Practice

Rational Expressions

Quiz MCQ 4 **Concept Quiz · MCQ**

Which statement is safe when simplifying a compound fraction?:

Choose the correct option.

- A Multiply top and bottom by the same inner denominator
- B Add all denominators
- C Cancel terms across +
- D Drop all restrictions

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ **and state the original restrictions.**

Classified Practice

Rational Expressions

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

Quiz WRITTEN 7 Concept Quiz · Written**Simplify**

$$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y} \text{ in } x \text{ and } y.$$

Classified Practice

Rational Expressions

Quiz WRITTEN 8 **Concept Quiz · Written**

Explain why $(x^2-16)/(x-4)=x+4$ is not an identity for every real x .

Answer key

Use the question number shown on each page.

Q001 A	Q024 C	Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$	Q066 $\frac{x+2}{x+1}$
Q002 B	Q025 D	Q044 $\frac{3}{x+3}$	Q067 $\frac{3x-2}{x(x-1)}$
Q003 C	Q026 D	Q045 $\frac{1}{x}$	Q068 $\frac{2x+15}{(x-1)(x+6)}$
Q004 D	Q027 A	Q046 $-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$	Q069 C
Q005 A	Q028 B	Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$	Q070 $\frac{2x}{(x-2)(x+2)}$
Q006 B	Q029 C	Q048 $x-1$	Q071 $-\frac{2}{x(x-1)}$
Q007 $\frac{4b^3}{3a^2}$	Q030 $\frac{x-2}{3x+1}$	Q049 C	Q072 $\frac{2}{(x+1)(x+2)}$
Q008 $\frac{3y^2}{10x}$	Q031 $\frac{2x-5}{x+2}$	Q050 D	Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$
Q009 $\frac{3a}{4b^3}$	Q032 $\frac{x}{x-3}$	Q051 A	Q074 $\frac{3x-1}{x(x-1)}$
Q010 $\frac{1}{4ab^2}$	Q033 $\frac{x+2}{x-4}$	Q052 B	Q075 $\frac{2}{(x+2)(x+4)}$
Q011 B	Q034 $\frac{2(x-y)}{x(x+4y)}$	Q053 $\frac{1}{x+1}$	Q076 $\frac{3}{(x+2)(x+3)}$
Q012 A	Q035 $\frac{x^2+x+1}{x+3}$	Q054 A	Q077 B
Q013 A	Q036 $\frac{x}{x-3}$	Q055 B	Q078 $\frac{x^2+2x+4}{x+1}$
Q014 B	Q037 $\frac{x-4}{x(x-2)}$	Q056 C	Q079 $-\frac{x^2-4x+2}{2(x-2)}$
Q015 A	Q038 $\frac{(x-5)(x+3)}{x(x+4)}$	Q057 C	Q080 $\frac{x-1}{x-2}$
Q016 B	Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$	Q058 $\frac{x+2}{x(x-1)}$	Q081 $x+1$
Q017 C	Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$	Q059 $\frac{x}{(x+1)(x+2)}$	Q082 D
Q018 D	Q041 $\frac{x-3}{(x-2)(2x-1)}$	Q060 B	Q083 A
Q019 A	Q042 $\frac{3ab^2}{4x^2}$	Q061 C	Q084 B
Q020 B		Q062 D	Q085 C
Q021 C		Q063 A	
Q022 $\frac{3c^2}{a}$		Q064 B	
Q023 $\frac{4a^2c^3}{b^2}$		Q065 C	

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$ Q097 $1 - x$ Q098 $x + 1$ Q099 $\frac{x}{y}$ Q100 $-(x + 1)$ Q101 $\frac{2x}{x^2 + 1}$ Q102 $\frac{x + 2}{2x + 3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x + 2}{x + 1}$ Quiz WRITTEN 6 $x - 1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4$; removable hole at (4, 8)

Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

LHS $\longrightarrow$ RHS

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Classified Practice

Identities and Proofs

Q001 **Challenge** $2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. Find **a,b,c**.

Classified Practice

Identities and Proofs

Q002 Written

$$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$$

Q003 Written

$$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$$

Classified Practice

Identities and Proofs

Q004 MCQ

For the identity $x^2 + 6x + 5 = a(x + 2)^2 + b(x + 2) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (2, 2, -3)

B (0, 2, -3)

C (1, 2, -3)

D (1, 3, -3)

Q005 MCQ

For the identity $2x^2 + 9x + 8 = a(x + 3)^2 + b(x + 3) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (3, -3, -1)

B (1, -3, -1)

C (2, -3, -1)

D (2, -2, -1)

Classified Practice

Identities and Proofs

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x + 4)^2 + b(x + 4) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (4, 4, 1)

B (2, 4, 1)

C (3, 5, 1)

D (3, 4, 1)

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x + 5)^2 + b(x + 5) + c$, the ordered tuple (a, b, c) is:

Choose the correct option.

A (1, -5, 3)

B (2, -5, 3)

C (0, -5, 3)

D (1, -4, 3)

Classified Practice

Identities and Proofs

Q008 Written

$$a(x + 1) - b(x + 5) = 8x$$

Q009 MCQ

For the identity $a(x + 1) - b(x + 3) = x - 1$, the ordered tuple (a, b) is:

Choose the correct option.

A (2, 1)

B (3, 1)

C (1, 1)

D (2, 2)

Classified Practice

Identities and Proofs

Q010 MCQ

For the identity $a(x + 2) - b(x + 4) = 5x + 14$, the ordered tuple (a, b) is:

Choose the correct option.

A $(4, -2)$

B $(3, -2)$

C $(2, -2)$

D $(3, -1)$

Q011 MCQ

For the identity $a(x + 3) - b(x + 5) = x - 3$, the ordered tuple (a, b) is:

Choose the correct option.

A (5, 3)

B (3, 3)

C (4, 4)

D (4, 3)

Classified Practice

Identities and Proofs

Q012 MCQ

For the identity $a(x + 4) - b(x + 6) = 9x + 44$, the ordered tuple (a, b) is:

Choose the correct option.

A (6, -4)

B (4, -4)

C (5, -3)

D (5, -4)

Q013 Written

$$(x + a)(b + 3) = 4x + 12$$

Classified Practice

Identities and Proofs

Q014 MCQ

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

Choose the correct option.

A $(3, -1)$

B $(2, -1)$

C $(1, -1)$

D $(2, 0)$

Q015 MCQ

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

Choose the correct option.

A (4, 0)

B (2, 0)

C (3, 0)

D (3, 1)

Classified Practice

Identities and Proofs

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Choose the correct option.

A (5, 1)

B (3, 1)

C (4, 1)

D (4, 2)

Q017 Written

$$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$$

Classified Practice

Identities and Proofs

Q018 Written

$$(ax + b)(x + 1) = 4x^2 + cx + 1$$

Q019 Written

$$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$$

Classified Practice

Identities and Proofs

Q020 MCQ

For the identity $(ax - 1)(x + 1) + c = x^2 - 3$, the ordered tuple (a, c) is:

Choose the correct option.

A $(2, -2)$

B $(1, -2)$

C $(0, -2)$

D $(1, -1)$

Q021 MCQ

For the identity $(ax + 2)(x + 2) + c = 2x^2 + 6x + 3$, the ordered tuple (a, c) is:

Choose the correct option.

A (3, -1)

B (1, -1)

C (2, -1)

D (2, 0)

Classified Practice

Identities and Proofs

Q022 MCQ

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

Choose the correct option.

A (4, 0)

B (3, 0)

C (2, 0)

D (3, 1)

Q023 Written

$$\frac{3x+1}{(x+2)(x-3)} = \frac{a}{x+2} + \frac{b}{x-3}$$

Classified Practice

Identities and Proofs

Q024 Written

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Q025 Written

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Classified Practice

Identities and Proofs

Q026 Written

$$\frac{a}{x-1} + \frac{b}{x-5} = \frac{5x-1}{(x-1)(x-5)}$$

Q027 Written

$$\frac{3x - 8}{3x^2 + 5x - 2} = \frac{a}{x + 2} + \frac{b}{3x - 1}$$

Classified Practice

Identities and Proofs

Q028 Written

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3}, \quad t > 0$$

Q029 Written**Prove that** $(x + y)^2 - (x - y)^2 = 4xy$.

Classified Practice

Identities and Proofs

Q030 Written

Prove that $(2a + b)^2 - (a + 2b)^2 = 3(a^2 - b^2)$.

Q031 MCQ

For the identity $-(x - 2y + 1)^2 + (x + 2y + 1)^2 = 2y(4x + 4)$, the value of the simplified left-hand side is:

Choose the correct option.

A $8xy + 8y$

B $8xy + 8y + 1$

C $8xy + 8y - 1$

D $-8xy - 8y$

Classified Practice

Identities and Proofs

Q032 MCQ

For the identity $-(x - 3y + 2)^2 + (x + 3y + 2)^2 = 3y(4x + 8)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $12xy + 24y + 1$
 - B $12xy + 24y - 1$
 - C $-12xy - 24y$
 - D $12xy + 24y$
-
-
-

Q033 MCQ

For the identity $-(x - y + 3)^2 + (x + y + 3)^2 = y(4x + 12)$, the value of the simplified left-hand side is:

Choose the correct option.

A $4xy + 12y + 1$

B $4xy + 12y - 1$

C $-4xy - 12y$

D $4xy + 12y$

Classified Practice

Identities and Proofs

Q034 MCQ

For the identity $-(x - 2y + 4)^2 + (x + 2y + 4)^2 = 2y(4x + 16)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $8xy + 32y$
- B $8xy + 32y + 1$
- C $8xy + 32y - 1$
- D $-8xy - 32y$

Q035 Written**Prove that** $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$.

Classified Practice

Identities and Proofs

Q036 MCQ

For the identity $(x - 2y + 1)^2 + (x + 2y + 1)^2 = 8y^2 + 2(x + 1)^2$, the value of the simplified left-hand side is:

Choose the correct option.

- A $2x^2 + 4x + 8y^2 + 3$
- B $2x^2 + 4x + 8y^2 + 1$
- C $2x^2 + 4x + 8y^2 + 2$
- D $-2x^2 - 4x - 8y^2 - 2$

Q037 MCQ

For the identity $(x - 3y + 2)^2 + (x + 3y + 2)^2 = 18y^2 + 2(x + 2)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2 + 8x + 18y^2 + 9$

B $2x^2 + 8x + 18y^2 + 7$

C $2x^2 + 8x + 18y^2 + 8$

D $-2x^2 - 8x - 18y^2 - 8$

Classified Practice

Identities and Proofs

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Choose the correct option.

- A $2x^2 + 12x + 2y^2 + 19$
- B $2x^2 + 12x + 2y^2 + 17$
- C $2x^2 + 12x + 2y^2 + 18$
- D $-2x^2 - 12x - 2y^2 - 18$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Choose the correct option.

A $2x^2 + 16x + 8y^2 + 33$

B $2x^2 + 16x + 8y^2 + 31$

C $-2x^2 - 16x - 8y^2 - 32$

D $2x^2 + 16x + 8y^2 + 32$

Classified Practice

Identities and Proofs

Q040 Written

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.

Q041 MCQ

For the identity $-(x - 2y + 3)^2 + (x + 2y - 1)^2 = (8x + 8)(y - 1)$, the value of the simplified left-hand side is:

Choose the correct option.

A $8xy - 8x + 8y - 7$

B $8xy - 8x + 8y - 9$

C $-8xy + 8x - 8y + 8$

D $8xy - 8x + 8y - 8$

Classified Practice

Identities and Proofs

Q042 MCQ

For the identity $-(2x - 3y + 5)^2 + (2x + 3y - 1)^2 = (24x + 24)(y - 1)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $24xy - 24x + 24y - 23$
- B $24xy - 24x + 24y - 25$
- C $24xy - 24x + 24y - 24$
- D $-24xy + 24x - 24y + 24$

Q043 Written**Prove that** $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

Classified Practice

Identities and Proofs

Q044 Written

Prove that $(a^2 - ab + b^2)(a^2 + ab + b^2) = (a^2 + b^2)^2 - (ab)^2$.

Q045 Written

Prove that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.

Classified Practice

Identities and Proofs

Q046 Written

Prove that $(x^2 + 1)(y^2 + 1) = (xy + 1)^2 + (x - y)^2$.

Q047 Written

Prove that $(a^2 - b^2)(c^2 - d^2) = (ac - bd)^2 - (ad - bc)^2$.

Classified Practice

Identities and Proofs

Q048 Written

Prove that $(a + b)^3 - 3ab(a + b) = a^3 + b^3$.

Q049 Written**Prove that** $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$.

Classified Practice

Identities and Proofs

Q050 Challenge

For the integrated identity $(x^2 + 1)(-(x - 2y + 1)^2 + (x + 2y + 1)^2) = 2y(4x + 4)(x^2 + 1)$, the value of the simplified left-hand side is:

Choose the correct option.

- A $8x^3y + 8x^2y + 8xy + 8y + 1$
- B $8x^3y + 8x^2y + 8xy + 8y$
- C $8x^3y + 8x^2y + 8xy + 8y - 1$
- D $-8x^3y - 8x^2y - 8xy - 8y$

Classified Practice

Identities and Proofs

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Q052 Written**Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.**

Classified Practice

Identities and Proofs

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2xy + 5$

B $-2xy + 3$

C $-2(xy - 2)$

D $2(xy - 2)$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 5)$

B $-2(xy - 4)$

C $2xy - 9$

D $-2xy + 9$

Classified Practice

Identities and Proofs

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 1)$

B $-2xy$

C $-2xy + 1$

D $2xy - 1$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $-2(xy - 8)$

B $-2xy + 17$

C $-2xy + 15$

D $2(xy - 8)$

Classified Practice

Identities and Proofs

Q057 Written

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2(xy + 1)$

B $2xy + 1$

C $2xy$

D $-2xy - 1$

Classified Practice

Identities and Proofs

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2xy + 5$

B $2(xy + 2)$

C $2xy + 3$

D $-2(xy + 2)$

Q060 MCQ**Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:****Choose the correct option.**

A $2xy + 5$

B $2xy + 3$

C $2(xy + 2)$

D $-2(xy + 2)$

Classified Practice

Identities and Proofs

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Choose the correct option.

A $2(xy + 5)$

B $2(xy + 4)$

C $-2xy - 9$

D $2xy + 9$

Q062 Written**Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.**

Classified Practice

Identities and Proofs

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Classified Practice

Identities and Proofs

Q065 **Written**

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Classified Practice

Identities and Proofs

Q067 Written

Under the condition $a + b + c = 0$, $abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Classified Practice

Identities and Proofs

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Choose the correct option.

A $-2xy + 11$

B $-2xy + 9$

C $-2(xy - 5)$

D $2(xy - 5)$

Classified Practice

Identities and Proofs

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}.$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $-\frac{cd}{c^2 + d^2}$
- D $\frac{cd}{c^2 + d^2}$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $\frac{cd}{c^2 + d^2}$

D $-\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $-\frac{cd}{c^2 + d^2}$
- D $\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q074 **Challenge**

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c^2 + cd + d^2}{c^2 + d^2}$

B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$

C $-\frac{cd}{c^2 + d^2}$

D $\frac{cd}{c^2 + d^2}$

Classified Practice

Identities and Proofs

Q075 **Challenge****Under the condition $a/b = c/d$, the required expression is equal to:****Choose the correct option.**

- A $\frac{c^2 + cd + d^2}{c^2 + d^2}$
- B $-\frac{c^2 - cd + d^2}{c^2 + d^2}$
- C $\frac{cd}{c^2 + d^2}$
- D $-\frac{cd}{c^2 + d^2}$
-
-
-

Classified Practice

Identities and Proofs

Q076 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Classified Practice

Identities and Proofs

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{a^2 + c^2}{(a - c)(a + c)}$

B $\frac{2a^2}{(a - c)(a + c)}$

C $\frac{2c^2}{(a - c)(a + c)}$

D $-\frac{a^2 + c^2}{(a - c)(a + c)}$

Classified Practice

Identities and Proofs

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{2a^2}{(a-c)(a+c)}$

B $\frac{2c^2}{(a-c)(a+c)}$

C $-\frac{a^2 + c^2}{(a-c)(a+c)}$

D $\frac{a^2 + c^2}{(a-c)(a+c)}$

Classified Practice

Identities and Proofs

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{2a^2}{(a-c)(a+c)}$

B $\frac{2c^2}{(a-c)(a+c)}$

C $-\frac{a^2 + c^2}{(a-c)(a+c)}$

D $\frac{a^2 + c^2}{(a-c)(a+c)}$

Classified Practice

Identities and Proofs

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{a^2 + c^2}{(a - c)(a + c)}$

B $\frac{2a^2}{(a - c)(a + c)}$

C $\frac{2c^2}{(a - c)(a + c)}$

D $-\frac{a^2 + c^2}{(a - c)(a + c)}$

Classified Practice

Identities and Proofs

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{2c}{c-d}$
B $\frac{c+d}{c-d}$
C $\frac{2d}{c-d}$
D $-\frac{c+d}{c-d}$

Classified Practice

Identities and Proofs

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{2c}{c-d}$
B $\frac{c+d}{c-d}$
C $\frac{2d}{c-d}$
D $-\frac{c+d}{c-d}$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c+d}{c-d}$

B $\frac{2c}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

Classified Practice

Identities and Proofs

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

A $\frac{c+d}{c-d}$

B $\frac{2c}{c-d}$

C $\frac{2d}{c-d}$

D $-\frac{c+d}{c-d}$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Classified Practice

Identities and Proofs

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Choose the correct option.

- A $\frac{2a - c}{2b - d}$
- B $\frac{2a + 2b - c - d}{2b - d}$
- C $\frac{2a - 2b - c + d}{2b - d}$
- D $-\frac{2a - c}{2b - d}$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Classified Practice

Identities and Proofs

Q089 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Q090 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$.

Classified Practice

Identities and Proofs

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+y+z}{a+b+c} = \frac{x}{a}$.

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$.

Classified Practice

Identities and Proofs

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Classified Practice

Identities and Proofs

Q095 Written

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Classified Practice

Identities and Proofs

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0, \frac{x^2+5y^2}{x^2+xy+y^2}$.

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Classified Practice

Identities and Proofs

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x^2 + 4x + 1 = (x + 1)(ax + b) + c$, then (a, b, c) is... Which of the following is correct?:

Choose the correct option.

A $(1, 3, -2)$

B $(1, 2, -1)$

C $(2, 3, -2)$

D $(1, -3, 2)$

Quiz MCQ 2 Concept Quiz · MCQ

For the identity $(x^2 - 2x + 2)(x^2 + 2x + 2) = -4x^2 + (x^2 + 2)^2$, the value of the simplified left-hand side is:

Choose the correct option.

- A $x^4 + 4$
 - B $x^4 + 5$
 - C $x^4 + 3$
 - D $-x^4 - 4$
-
-
-

Classified Practice

Identities and Proofs

Quiz MCQ 3 **Concept Quiz · MCQ**

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Choose the correct option.

A $3b(b + c)$

B $3b^2 + 3bc + 1$

C $3b^2 + 3bc - 1$

D $-3b(b + c)$

Quiz MCQ 4 Concept Quiz · MCQ

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Choose the correct option.

- A $\frac{5}{2}$
- B $\frac{3}{2}$
- C $\frac{1}{2}$
- D $-\frac{3}{2}$

Classified Practice

Identities and Proofs

Quiz WRITTEN 5 Concept Quiz · Written**Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.**

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $(x^2 - 3x + 4)(x^2 + 3x + 4) = -9x^2 + (x^2 + 4)^2$.

Classified Practice

Identities and Proofs

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Answer key

Use the question number shown on each page.

Q001 $a = 2, b = 1, c = 0$

Q002 $a = 1, b = -2, c = 7$

Q003 $a = 1, b = 1, c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10, b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3, b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2, b = 3, c = 1$

Q018 $a = 4, b = 1, c = 5$

Q019 $a = 2, b = -1, c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1, b = 2$

Q024 $a = 2, b = -1$

Q025 $a = \frac{2}{3}, b = -\frac{1}{3}$

Q026 $a = -1, b = 6$

Q027 $a = 2, b = -3$

Q028 $a = 4, b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$

Q068 C

Q069 $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$

Q087 A

Q088 $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$

Q089 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q090 $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$

Q091 $\frac{x+y+z}{a+b+c} = \frac{x}{a}$

Q092 $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A**Quiz MCQ 2** A**Quiz MCQ 3** A**Quiz MCQ 4** B**Quiz WRITTEN 5** $a = 1, b = -1, c = 3$ **Quiz WRITTEN 6** $x^4 - x^2 + 16$ **Quiz WRITTEN 7** $4b^2 + 4bc$ **Quiz WRITTEN 8** $(ax + by + cz)^2$

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Classified Practice

Inequalities

Q001 Written

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Q002 MCQ**If $x > y$, prove that $3x + y > 2(x + y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

Classified Practice

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y + 1$

B $x - y - 1$

C $x - y$

D $-x + y$

Q004 MCQ**If $x > y$, prove that $3(x + y) > 2(x + 2y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

Classified Practice

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

Q006 MCQ**If $x > y$, prove that $3x + 5y > 2(x + 3y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y + 1$

B $x - y$

C $x - y - 1$

D $-x + y$

Classified Practice

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Classified Practice

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $ac - ad - bc + bd + 1$

B $ac - ad - bc + bd - 1$

C $-(a - b)(c - d)$

D $(a - b)(c - d)$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $ac - ad - 2bc + 2bd + 1$

B $ac - ad - 2bc + 2bd - 1$

C $(a - 2b)(c - d)$

D $-(a - 2b)(c - d)$

Classified Practice

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $(a - 3b)(c - d)$

B $ac - ad - 3bc + 3bd + 1$

C $ac - ad - 3bc + 3bd - 1$

D $-(a - 3b)(c - d)$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Classified Practice

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Classified Practice

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Classified Practice

Inequalities

Q017 Written

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Classified Practice

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $2ax - 2ay - 2bx + 2by + 1$

B $2ax - 2ay - 2bx + 2by - 1$

C $-2(a - b)(x - y)$

D $2(a - b)(x - y)$

Classified Practice

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

- A $a^2 - 4ab + 4b^2 + 1$
- B $(a - 2b - 1)(a - 2b + 1)$
- C $-(a - 2b)^2$
- D $(a - 2b)^2$

Classified Practice

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $a^2 - 6ab + 9b^2 + 1$
B $(a - 3b - 1)(a - 3b + 1)$
C $(a - 3b)^2$
D $-(a - 3b)^2$

Q022 MCQ**Under a,b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:****Choose the correct option.**

A $a^2 - 8ab + 16b^2 + 1$

B $(a - 4b)^2$

C $(a - 4b - 1)(a - 4b + 1)$

D $-(a - 4b)^2$

Classified Practice

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

Q024 MCQ

Under a,b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 12ab + 36b^2 + 1$

B $(a - 6b - 1)(a - 6b + 1)$

C $(a - 6b)^2$

D $-(a - 6b)^2$

Classified Practice

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 14ab + 49b^2 + 1$

B $(a - 7b)^2$

C $(a - 7b - 1)(a - 7b + 1)$

D $-(a - 7b)^2$

Q026 Written**Prove that** $a^2 + 5b^2 \geq 4ab$.

Classified Practice

Inequalities

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.

Q028 Written**Prove that** $2(x^2 + y^2) \geq (x + y)^2$.

Classified Practice

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.

Q030 Written**Prove that** $(2a + 3b)^2 \geq 12ab$.

Classified Practice

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $(2x - y)^2$

B $4x^2 - 4xy + y^2 + 1$

C $(2x - y - 1)(2x - y + 1)$

D $-(2x - y)^2$

Q032 MCQ**Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:****Choose the correct option.**

- A $9x^2 - 6xy + y^2 + 1$
 - B $(3x - y - 1)(3x - y + 1)$
 - C $(3x - y)^2$
 - D $-(3x - y)^2$
-
-
-

Classified Practice

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $16x^2 - 8xy + y^2 + 1$

B $(4x - y - 1)(4x - y + 1)$

C $-(4x - y)^2$

D $(4x - y)^2$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $25x^2 - 10xy + y^2 + 1$

B $(5x - y - 1)(5x - y + 1)$

C $-(5x - y)^2$

D $(5x - y)^2$

Classified Practice

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $(6x - y)^2$

B $36x^2 - 12xy + y^2 + 1$

C $(6x - y - 1)(6x - y + 1)$

D $-(6x - y)^2$

Q036 MCQ**Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:****Choose the correct option.**

A $x^2 - 2xy + y^2 + 1$

B $(x - y - 1)(x - y + 1)$

C $(x - y)^2$

D $-(x - y)^2$

Classified Practice

Inequalities

Q037 **Written****Prove that** $x^2 + y^2 \geq 2(x - y - 1)$.

Q038 Written**Prove that** $x^2 + 2 > 2x$.

Classified Practice

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.

Q040 Written**Prove that** $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Classified Practice

Inequalities

Q041 **Written****Prove that** $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.

Classified Practice

Inequalities

Q042 **Challenge**

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Classified Practice

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Classified Practice

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a-b)^2}{4}$

B $\frac{a^2 - 2ab + b^2 + 4}{4}$

C $\frac{(a-b-2)(a-b+2)}{4}$

D $-\frac{(a-b)^2}{4}$

Classified Practice

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 2b)^2}{4}$

B $\frac{a^2 - 4ab + 4b^2 + 4}{4}$

C $\frac{(a - 2b - 2)(a - 2b + 2)}{4}$

D $-\frac{(a - 2b)^2}{4}$

Classified Practice

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{a^2 - 6ab + 9b^2 + 4}{4}$
- B $\frac{(a - 3b - 2)(a - 3b + 2)}{4}$
- C $\frac{(a - 3b)^2}{4}$
- D $-\frac{(a - 3b)^2}{4}$

Classified Practice

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 4b)^2}{4}$

B $\frac{a^2 - 8ab + 16b^2 + 4}{4}$

C $\frac{(a - 4b - 2)(a - 4b + 2)}{4}$

D $-\frac{(a - 4b)^2}{4}$

Classified Practice

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 5b)^2}{4}$

B $\frac{a^2 - 10ab + 25b^2 + 4}{4}$

C $\frac{(a - 5b - 2)(a - 5b + 2)}{4}$

D $-\frac{(a - 5b)^2}{4}$

Classified Practice

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{a^2 - 12ab + 36b^2 + 4}{4}$

B $\frac{(a - 6b)^2}{4}$

C $\frac{(a - 6b - 2)(a - 6b + 2)}{4}$

D $-\frac{(a - 6b)^2}{4}$

Classified Practice

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 7b)^2}{4}$

B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$

C $\frac{(a - 7b - 2)(a - 7b + 2)}{4}$

D $-\frac{(a - 7b)^2}{4}$

Classified Practice

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 8b)^2}{4}$

B $\frac{a^2 - 16ab + 64b^2 + 4}{4}$

C $\frac{(a - 8b - 2)(a - 8b + 2)}{4}$

D $-\frac{(a - 8b)^2}{4}$

Classified Practice

Inequalities

Q052 Written

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.

Q053 Written**Prove that** $a + 2 \geq \sqrt{3 + 6a}$.

Classified Practice

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$.

Classified Practice

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $2\sqrt{a}\sqrt{b} + 1$

B $2\sqrt{a}\sqrt{b}$

C $2\sqrt{a}\sqrt{b} - 1$

D $-2\sqrt{a}\sqrt{b}$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $4\sqrt{a}\sqrt{b}$

B $4\sqrt{a}\sqrt{b} + 1$

C $4\sqrt{a}\sqrt{b} - 1$

D $-4\sqrt{a}\sqrt{b}$

Classified Practice

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $6\sqrt{a}\sqrt{b} + 1$

B $6\sqrt{a}\sqrt{b} - 1$

C $6\sqrt{a}\sqrt{b}$

D $-6\sqrt{a}\sqrt{b}$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $8\sqrt{a}\sqrt{b} + 1$

B $8\sqrt{a}\sqrt{b}$

C $8\sqrt{a}\sqrt{b} - 1$

D $-8\sqrt{a}\sqrt{b}$

Classified Practice

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Q061 Written**Prove that** $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.

Classified Practice

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

Q063 Written**Prove that** $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$.

Classified Practice

Inequalities

Q064 Written

Prove that $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.

Q065 Written

If $a > b > 0$, prove that $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.

Classified Practice

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Classified Practice

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Q068 Written**Prove that** $\frac{b}{3a} + \frac{12a}{b} \geq 4$.

Classified Practice

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2 + b^2}{ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{(a - b)^2}{ab}$

B $\frac{a^2 - ab + b^2}{ab}$

C $\frac{a^2 - 3ab + b^2}{ab}$

D $-\frac{(a - b)^2}{ab}$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{4a^2 - 2ab + b^2}{2ab}$

B $\frac{4a^2 - 6ab + b^2}{2ab}$

C $\frac{(2a - b)^2}{2ab}$

D $-\frac{(2a - b)^2}{2ab}$

Classified Practice

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{9a^2 - 3ab + b^2}{3ab}$

B $\frac{(3a - b)^2}{3ab}$

C $\frac{9a^2 - 9ab + b^2}{3ab}$

D $-\frac{(3a - b)^2}{3ab}$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{(4a - b)^2}{4ab}$

B $\frac{16a^2 - 4ab + b^2}{4ab}$

C $\frac{16a^2 - 12ab + b^2}{4ab}$

D $-\frac{(4a - b)^2}{4ab}$

Classified Practice

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{25a^2 - 5ab + b^2}{5ab}$
- B $\frac{25a^2 - 15ab + b^2}{5ab}$
- C $-\frac{(5a - b)^2}{5ab}$
- D $\frac{(5a - b)^2}{5ab}$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{36a^2 - 6ab + b^2}{6ab}$

B $\frac{36a^2 - 18ab + b^2}{6ab}$

C $-\frac{(6a - b)^2}{6ab}$

D $\frac{(6a - b)^2}{6ab}$

Classified Practice

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{49a^2 - 7ab + b^2}{7ab}$
- B $\frac{49a^2 - 21ab + b^2}{7ab}$
- C $\frac{(7a - b)^2}{7ab}$
- D $-\frac{(7a - b)^2}{7ab}$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{64a^2 - 8ab + b^2}{8ab}$

B $\frac{64a^2 - 24ab + b^2}{8ab}$

C $-\frac{(8a - b)^2}{8ab}$

D $\frac{(8a - b)^2}{8ab}$

Classified Practice

Inequalities

Q077 Written

Prove that $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.

Q078 Written**Prove that** $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.

Classified Practice

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.

Q080 Written**Prove that** $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.

Classified Practice

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{a^2 - ab + b^2}{ab}$

B $\frac{a^2 - 3ab + b^2}{ab}$

C $-\frac{(a-b)^2}{ab}$

D $\frac{(a-b)^2}{ab}$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

Classified Practice

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{9a^2 - 17ab + 9b^2}{ab}$

B $\frac{9a^2 - 19ab + 9b^2}{ab}$

C $\frac{9(a-b)^2}{ab}$

D $-\frac{9(a-b)^2}{ab}$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{16a^2 - 31ab + 16b^2}{ab}$

B $\frac{16a^2 - 33ab + 16b^2}{ab}$

C $-\frac{16(a-b)^2}{ab}$

D $\frac{16(a-b)^2}{ab}$

Classified Practice

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.

Q086 Written**Prove that $a + \frac{4}{a} \geq 4$.**

Classified Practice

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.

Q088 Written**Prove that** $a + b + \frac{9}{a+b} \geq 6$.

Classified Practice

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Classified Practice

Inequalities

Q090 **Challenge**

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Classified Practice

Inequalities

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

Quiz MCQ 2 Concept Quiz · MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

Classified Practice

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 7b)^2}{4}$
- B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$
- C $\frac{(a - 7b - 2)(a - 7b + 2)}{4}$
- D $-\frac{(a - 7b)^2}{4}$

Classified Practice

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

Quiz WRITTEN 5 **Concept Quiz · Written**

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Classified Practice

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written**Under x, y real, prove that $x^2 + y^2 \geq 2xy$.**

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Classified Practice

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2